Annotation Convention, Not Architecture, Limits Cross-Facility Detection of Electric Vehicle Battery Components

Komiljon Kosimov and Shiv Ashutosh Katiyar
Department of Mechanical Engineering, University of Birmingham, Birmingham B15 2TT, United Kingdom
kxk445@student.bham.ac.uk, s.a.katiyar@bham.ac.uk

Abstract—Automated disassembly of end-of-life electric vehicle (EV) battery packs depends on a detector that locates modules and busbars across the pack types a recycling line actually receives. Published detectors for this task are trained and validated at a single facility, and the accuracy they report is shown here to be a poor predictor of behaviour elsewhere. This paper presents a cross-facility benchmark of 225 images and 780 instances, assembled from 13 independent public sources spanning ten pack families and verified free of train-test leakage by perceptual hashing. On it, a single-source specialist falls from 0.818 to 0.277 mAP@50. Disaggregating by source shows the loss is concentrated rather than spread: per-source module accuracy spans 0.995 to 0.043, and the collapse occurs on one source whose annotation convention differs from the consensus. This is developed into a screening procedure that identifies convention-divergent sources from a single training run using a robust outlier criterion, and excluding the one flagged source raises module mAP@50 from 0.547 to 0.774 with no additional data. A training-free variant, computed from annotation geometry over 4,760 label files, flags the same source with no training at all and passes a positive control in which a consensus source relabelled to a sub-component convention is flagged while every genuine consensus source is not. A detection transformer on a self-supervised pretrained backbone is shown to confer no advantage on convention-consistent sources (0.771 against 0.774) while degrading far more gracefully under convention shift (0.680 against 0.547). Four negative results constrain the design space. The benchmark and evaluation code are released.

Index Terms—object detection, dataset quality, annotation consistency, domain generalisation, cross-dataset evaluation, electric vehicle batteries, circular economy

I. Introduction

Electric vehicles are reaching end of life in large numbers, and the packs they carry are simultaneously a hazard and a source of critical raw materials. Lithium, cobalt and nickel are all classified by the European Commission as critical raw materials with elevated supply risk [1], [2], and European recycling capacity demand is projected to reach 170 GWh by 2035 [3]. Disassembly of these packs remains predominantly manual. The operation is labour intensive, slow, and exposes workers to high voltage and to the risk of thermal runaway, which motivates a substantial and growing research effort on robotic disassembly [4], [6].

Every such system begins with perception. Before a manipulator can unfasten, grasp or lift a component, the battery modules and the busbars that connect them must be located in the image. The established approach is a supervised object detector trained on imagery collected at one facility or from one experimental rig. This practice conceals a problem specific to the application. A recycling line does not receive a single pack type; it receives whatever arrives on a given day, from different manufacturers, in different states of disassembly, under whatever illumination the building provides. Accuracy measured on the distribution a detector was trained on therefore carries limited information about the next pack through the door.

To quantify this, a cross-facility benchmark for EV battery component detection is required, and none exists. Every study surveyed in Section II collected and annotated its own imagery, and none released it. To address this gap, this paper assembles a corpus of 4,425 training images from 13 independent public sources spanning ten EV pack families, constructs a held-out cross-facility benchmark of 225 images and 780 instances, and evaluates detectors on it under a single harmonised protocol.

The principal finding is that the dominant failure mode is not image difficulty and not model capacity. It is disagreement between sources about what the target class denotes. Per-source module accuracy spans 0.995 to 0.043, and inspection confirms that the collapsing source annotates small internal sub-components as modules while every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently; the collapse is a disagreement about the target, not a failure of perception.

This paper makes the following contributions:

1) A cross-facility benchmark for EV battery module and busbar detection, comprising 225 images and 780 instances drawn from 13 sources and ten pack families, verified free of train-test leakage by perceptual hashing and released with a per-source provenance and licence record (Section III).

Benchmark definition, per-source provenance and licence record, audit tooling, evaluation code and detector weights: https://github.com/Komil-jon/ev-battery-vision-pipeline

2) A convention-consistency screening procedure that identifies annotation-divergent sources from a single merged-corpus training run, using a robust outlier criterion on per-source disaggregated accuracy. Excluding the one source flagged by the proposed procedure raises module mAP@50 from 0.547 to 0.774 with no additional data and no additional training budget (Sections IV-C and V-B).
3) Quantification of the single-source evaluation gap for this task: 0.818 mAP@50 in domain against 0.277 across facilities, a relative loss of 66% (Section V-A).
4) A characterisation of what a self-supervised pretrained backbone supplies in this setting. No advantage is observed on convention-consistent sources (0.771 against 0.774); the entire aggregate advantage arises from graceful degradation under annotation shift (Section V-E).
5) A training-free variant of the screen, computed from annotation geometry alone, which flags the same source with no training run (Sections IV-D and V-C).
6) Four negative results, covering open-vocabulary detection, automatic relabelling, naive multi-source merging and synthetic augmentation, which constrain the design space for this task (Section V-G).

Relation to prior work by the same authors. The single-source specialist evaluated here is the detector of [14], which addressed detection and condition assessment at one facility. That work could not test cross-facility transfer, which is the limitation this paper addresses. The corpus, the benchmark, both screening procedures and all evaluation reported here are new.

II. Related Work

A. Perception for battery disassembly

Zang et al. [4] surveyed robotic EV battery disassembly and concluded that the process remains predominantly manual, with perception and task uncertainty identified as the principal obstacles. Wegener et al. [5] demonstrated that pack designs vary sufficiently between manufacturers that fixed automation does not transfer, which is the practical reason perception must generalise. However, that study addressed mechanical disassembly sequencing and did not evaluate a perception component.

Most published perception work for this application targets fasteners rather than components. Screw detection has been addressed with Faster R-CNN combined with rotation edge similarity [7], with YOLOX-family detectors [8], and in a recent comparison of YOLOv11 against Faster R-CNN [9]. However, each of these studies reports accuracy on imagery from a single collection and therefore cannot establish transfer. Module-level and busbar-level perception is comparatively under-explored. Tan et al. [6] developed a collaborative robot that unfastens and collects screws by fusing vision with force and torque sensing, reporting above 90% unfastening efficiency; however, the vision component operates on a single fixed pack geometry. The closest prior work applies instance segmentation with point-cloud registration to grasp busbars [10], but is restricted to one module type.

A recurring theme is data scarcity. Every study named above collected and hand-annotated its own imagery, 1,719 images in the case of [8], and none released it. [9] resorts to laptop screws as a proxy domain and states that the field is hampered by a lack of public datasets. That absence motivates the benchmark presented here.

B. Cross-dataset evaluation and label quality

Domain shift is a well-documented failure mode in visual recognition [11], [12]: models that appear strong on their own test split degrade sharply on data collected under a different protocol. However, these benchmarks vary the imaging distribution while holding the annotation protocol fixed, whereas the corpus considered here varies both. Northcutt et al. [13] demonstrated that pervasive label errors in established test sets destabilise benchmark rankings, but treated label error as noise about an agreed target rather than as systematic disagreement about what the target is. The distinction matters for merged corpora, and it is the distinction the proposed screening procedure operationalises. Penquitt et al. [18] developed a modular framework combining automatic label-error detection with crowd-sourced verification, recovering 18% missing or inaccurate labels in the KITTI pedestrian class. However, that framework corrects errors relative to a standard agreed within a single dataset, whereas the sources considered here disagree about what the class denotes, which no within-convention correction can reconcile. Abou Akar et al. [19] studied multi-source training for industrial detection using real, rendered and generatively synthesised imagery. However, those sources share one annotation standard by construction, since synthetic data carries its ground truth.

C. Self-supervised representations for detection

Benchmark design for industrial vision has precedent: MVTec AD [15] established that a carefully constructed benchmark can redirect a sub-field, though it was collected by one group under controlled conditions and so exhibits no inter-source annotation disagreement. Zhou et al. [17] surveyed domain generalisation; however, that taxonomy presumes a fixed label space across domains, the assumption shown here to fail on assembled corpora. Self-supervised backbones such as DINOv2 [20] learn features that transfer without task-specific fine-tuning, and detection transformers [16] built on them [21] are reported to generalise better than detectors trained from scratch in low-data regimes. However, that claim has been established on natural imagery with consistent annotation. Whether it holds on the cluttered, specular imagery of battery internals, and under annotation shift, is evaluated in Section V-E.

TABLE I
Corpus sources (13 total, ten pack families)

Source	Images	Pack family
ev-battery-comp-det.	1,045	multi-manufacturer
ev-battery-components	680	Audi / BMW / Chevrolet
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single laboratory rig
battery_comp, ue_d1	517	Nissan Leaf, close-ups
Others (eight sources)	1,265	mixed

To the best of our knowledge, no public cross-facility benchmark for EV battery component detection exists, and no prior study has quantified the contribution of annotation-convention divergence to cross-source detection accuracy. Therefore, this research presents both.

III. Corpus and Benchmark

A. Sources

A corpus of 4,425 training images was assembled from 13 independent public sources spanning ten EV pack families, annotated with two classes, module and busbar (Table I). The images and the polygon annotations originate with those sources; the contribution made here is the assembly, the deduplication, the audit and the benchmark construction, not the annotation. Licences differ across sources and one is non-commercial, so per-source provenance is recorded and released so that any single source can be excluded by a downstream user.

B. Deduplication and leakage control

Reported accuracy is informative only against a benchmark that does not overlap the training data. All 4,440 candidate images were therefore compared by difference hashing (dHash) at a Hamming distance threshold of 6, a value selected because it separates augmentation-derived near-duplicates from genuinely distinct frames on this corpus. Fifteen same-source near-duplicates were removed, and a cross-source pass at a looser threshold returned zero duplicates, which confirms that the sources are genuinely independent rather than republications of one another.

This control proved necessary rather than precautionary. One candidate source, a 712-image public collection covering 19 vehicle types, republishes imagery from another source already present in the corpus: 78 images were byte-identical duplicates of training images. Merging without verification would have placed training images directly into the benchmark.

C. Annotation audit

All 4,425 training images were audited for label and image quality. Of these, 23.5% carry no labels, 12.3% score low on a Laplacian-variance blur metric, 20.4% are greyscale, and none contain degenerate boxes.

Two findings changed how the corpus was treated. Firstly, the blur and greyscale flags are largely false alarms. The lowest-scoring images are smooth, closed pack lids, for which a Laplacian-variance metric reads flat metal as blurred, and the greyscale images originate from two monochrome sources. Both add useful variation, so deletion on the metric alone would degrade the corpus. Secondly, the harmful category is unlabelled images that nonetheless contain visible components. A proportion of the 23.5% is legitimate background, which suppresses false positives, but inspection confirmed open packs with clearly visible busbar assemblies carrying no annotations. Such frames teach the detector that busbars are background and are a plausible contributor to the weak busbar accuracy reported in Section V-E.

The audit also established that 16,945 of the 19,522 annotations are polygon masks with between 5 and 151 vertices rather than boxes. These are converted to enclosing boxes for detection training and evaluation. It should be noted that an early version of the evaluation pipeline silently discarded every polygon-format label, trained on 20% of the annotations, and produced an inflated score that appeared plausible. The failure is easy to introduce and difficult to detect, and ground-truth instance counts should be reconciled across pipelines as a matter of routine.

D. The cross-facility benchmark

A cross-facility benchmark of 225 images containing 780 instances was drawn from all sources and held out entirely from training. A convention-consistent subset of 177 images, obtained by applying the screening procedure of Section IV-C, is reported alongside it throughout.

IV. Method

A. Detectors and training configurations

Three configurations are compared.

Single-source specialist. YOLOv8n trained only on the single-laboratory source at 768 px, following the conventional approach in this literature.

Multi-source YOLO11n. YOLO11n [22], 2.6 M parameters, trained from COCO initialisation on all 13 sources for 150 epochs at 640 px with stochastic gradient descent.

Multi-source RF-DETR. RF-DETR-Nano [21], a real-time detection transformer initialised from the published RF-DETR-Nano checkpoint, whose backbone derives from self-supervised DINOv2 pretraining [20], fine-tuned end to end on the same data. All 30.6 M parameters are updated; the backbone is not frozen. The resolution and epoch budget are the architecture defaults; the consequence for comparability is stated in Section VI.

B. Harmonised evaluation protocol

Confidence scores are not comparable across architectures, and mixing metric implementations produced a spurious result in early experiments conducted for this study. Every model is therefore scored with a single metric implementation at a common confidence threshold of 0.05, on identical images, against identical ground truth including polygon-derived boxes. The threshold of 0.05 was selected because it lies below the operating point of all three detectors and therefore does not truncate the precision-recall curve for any of them.

C. Convention-consistency screening

The proposed screening procedure identifies sources whose annotation convention diverges from the corpus consensus, using a single merged-corpus training run.

Let $\mathcal{S} = \{S_1, \ldots, S_n\}$ denote the sources and let a_i denote the class-specific mAP@50 obtained by the merged-corpus detector on the held-out images of S_i. A source is flagged as convention-divergent when its accuracy falls below a robust lower fence,

$$a_i < \mathrm{med}(\mathbf{a}) - k \cdot 1.4826 \cdot \mathrm{mad}(\mathbf{a}), \tag{1}$$

where $\mathbf{a} = (a_1, \ldots, a_n)$, med denotes the median, mad denotes the median absolute deviation, the constant 1.4826 rescales the median absolute deviation to a consistent estimator of the standard deviation under normality, and k is a sensitivity parameter. A value of $k = 3$ is used throughout, which is the conventional choice for outlier rejection and which is shown in Section V-B to separate the divergent source from the consensus by a wide margin on this corpus.

The median and the median absolute deviation are used in place of the mean and the standard deviation because a divergent source inflates both of the latter and thereby masks itself. The procedure requires no additional annotation, no additional training run beyond the merged-corpus model that is trained in any case, and no assumption about which source is divergent.

A flagged source is not discarded from the corpus. It is reported separately as an out-of-convention benchmark, on the grounds established in Section V-B that its labels encode a different task rather than a degraded version of the same one.

D. Training-free screening from annotation geometry

The procedure above requires a merged-corpus training run. A cheaper variant operates on the annotation files alone. A source annotating sub-components rather than pack-level modules produces many small boxes per image, whereas a source following the consensus convention produces few large ones. That signature is captured by a granularity index, defined as follows:

$$g_i = n_i / s_i, \tag{2}$$

where n_i denotes the mean number of module instances per image in source S_i and s_i denotes the median normalised module scale, taken as the square root of the enclosing-box area in normalised image coordinates. Scale enters as a square root so that the index is a count per unit length and is invariant to image resolution. Eq. 1 is applied to $\mathbf{g}$ with the inequality reversed, since divergence raises the index.

TABLE II
Cross-facility generalisation gap, mAP@50

Model	In-domain	Cross-facility
Single-source specialist	0.818	0.277
Multi-source YOLO11n	0.195	0.410
Multi-source RF-DETR	0.541	0.502

V. Results

A. The single-source evaluation gap

Table II quantifies the central problem. The specialist scores 0.818 mAP@50 on the single-source benchmark on which it was trained and validated, and 0.277 on the cross-facility benchmark, a relative loss of 66%. Its in-domain accuracy therefore does not predict deployment behaviour.

The converse also holds and is worth stating: the multi-source models score lower in domain than the specialist. Training for breadth costs peak accuracy on any single pack type, and which trade is preferable depends on whether a line receives one pack type or many. However, the magnitude of the drop for YOLO11n (0.195) exceeds what breadth alone explains. It is not a training failure. The in-domain benchmark is itself the source whose annotation convention diverges from the remainder of the corpus, a confound examined in Section V-B and stated as a limitation in Section VI.

B. Screening outcome and recovery

Per-source module accuracy for the merged-corpus detector is 0.995 on ue_rav4, 0.910 on bmw_i3, 0.873 on the largest source, 0.749 on ev-battery-components, and 0.043 on the single-laboratory source. Applying Eq. 1 with $k = 3$ yields a lower fence of 0.330. Fig. 1 shows that exactly one source falls below it, and that the margin is wide: the flagged source scores 0.287 below the fence, while the nearest consensus source scores 0.419 above it. The screen is therefore not sensitive to the precise value of k on this corpus; any k between 1.5 and 4.5 produces the identical partition.

Visual inspection of the flagged source confirms the diagnosis. It annotates small internal sub-components in extreme macro views as modules, whereas every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the collapse constitutes a disagreement about the target rather than a failure of perception.

The consequence is quantified in Fig. 2: excluding the flagged source raises YOLO11n module mAP@50 from 0.547 to 0.774, a recovery of 0.227, with no additional data and no additional training. Aggregate accuracy over sources with inconsistent conventions is therefore not a meaningful quantity.

C. Agreement of the training-free screen

Eq. 2 was evaluated over all 4,760 label files, using no images and no trained model. The divergent source

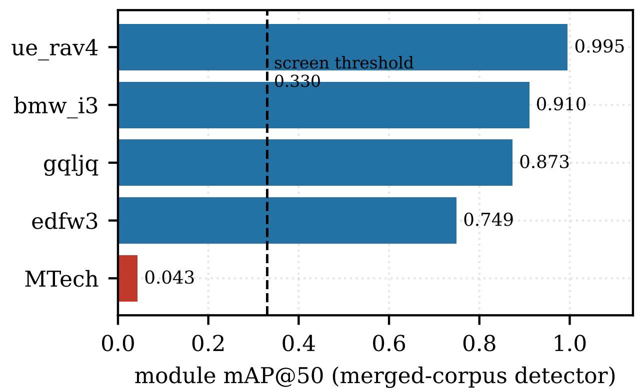

Fig. 1. Per-source module accuracy for the merged-corpus detector, with the robust lower fence of Eq. 1 at $k = 3$. One source falls below the fence by a wide margin.

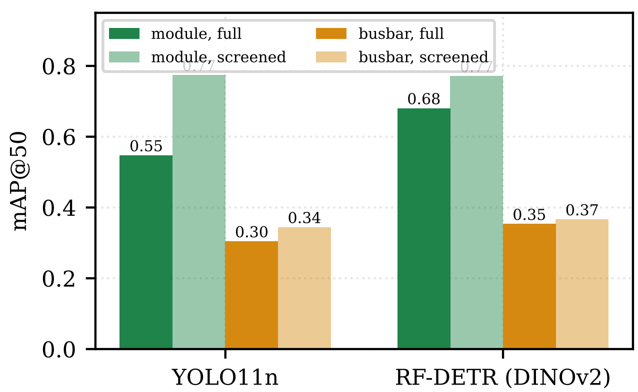

Fig. 2. Per-class accuracy on the full benchmark and on the screened subset. Screening recovers 0.227 module mAP@50 for YOLO11n and 0.091 for RF-DETR, and leaves busbar accuracy substantially unchanged.

attains a granularity index of 40.7, against 15.0 for the next highest source and a corpus median of 9.00. The robust criterion at $k = 3$ yields an upper fence of 29.02, and exactly one source exceeds it (Fig. 3). It is the same source flagged by the model-based procedure. The two screens are methodologically independent, one measuring detector behaviour on held-out images and the other only annotation geometry, so their agreement is a second line of evidence that the source encodes a different convention rather than harder imagery. A corpus may therefore be screened before a detector is trained. It should be noted that the index detects divergence in annotation granularity specifically; conventions disagreeing about class boundaries while preserving object scale would still require the model-based procedure.

D. Positive control

Both screens were applied to a corpus in which the divergent source was identified after the fact, so neither

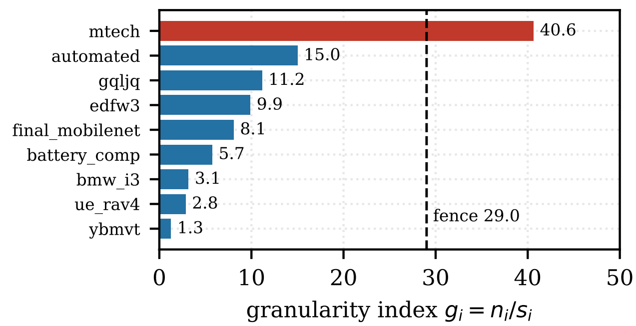

Fig. 3. Training-free convention screening. The granularity index of Eq. 2 per source, with the robust fence at $k = 3$. The flagged source is the one the model-based procedure flags, identified with no training run.

TABLE III
Detector comparison on the full cross-facility benchmark (225 images) and on the screened subset (177 images), mAP@50 unless stated

Model	Benchmark	@50	@50–95	module	busbar
Specialist	full	0.277	–	0.231	–
YOLO11n	full	0.410	0.256	0.547	0.304
	screened	0.544	–	0.774	0.344
RF-DETR	full	0.502	0.312	0.680	0.353
	screened	0.558	–	0.771	0.366

has been tested against a case whose answer is known in advance. To supply one, a consensus source was relabelled to a deliberately different convention, every pack-level module box being replaced by a 3×2 grid of inset sub-component boxes covering the same region. No image is altered and no busbar annotation is touched, so a screen responding to the manipulation responds to annotation convention alone. The manipulated source carries a granularity index of 11.1 before manipulation, inside the consensus band, and 185.8 after. The training-free screen flags it, continues to flag the real divergent source, and leaves every genuine consensus source below the fence, so the procedure recovers a divergence whose ground truth is known by construction without false positives elsewhere. This control validates the training-free screen; an equivalent control for the model-based procedure requires retraining on the manipulated corpus and is reported as future work.

E. Architecture comparison under annotation shift

Under the harmonised protocol, RF-DETR attains 0.502 mAP@50 against 0.410 for YOLO11n on the full benchmark, and leads on the stricter mAP@50–95 (0.312 against 0.256), as reported in Table III.

Taken alone that comparison is misleading. Table III also reports both models on the screened subset of 177 images, on which they are effectively tied: 0.558 against

0.544 overall, 0.771 against 0.774 on the module class, and 0.366 against 0.344 on the busbar class. The self-supervised pretrained backbone therefore confers no advantage on conventionally annotated packs.

The aggregate advantage arises entirely from behaviour under annotation shift. Reintroducing the flagged source reduces YOLO11n module accuracy from 0.774 to 0.547, a loss of 0.227, but reduces RF-DETR module accuracy only from 0.771 to 0.680, a loss of 0.091. The defensible claim is correspondingly narrow: a self-supervised pretrained backbone does not detect ordinary modules more accurately, but it degrades considerably more gracefully when the definition of the target moves. On a line receiving heterogeneous packs, graceful degradation is the property of practical value.

Busbar is the weaker class for every configuration, at 0.30 to 0.37. The cause is annotation coverage rather than model capability: only four of eight benchmark sources annotate busbars at all, and 61% of busbar instances originate from a single source. This is the same coverage problem identified for the module class, expressed in a different form.

F. Deployment cost and zero-shot transfer

Evaluated on 17 pack families absent from the corpus, drawn from an unlabelled public collection, the merged-corpus detector attains a mean detection rate of 0.76 and finds busbars in 16 of 17, with 1.00 on BMW i4 and Hyundai Ioniq against 0.27 on a Volvo truck pack, so module geometry transfers to unseen passenger-vehicle packs while unusual form factors remain the weak axis. Dynamic INT8 quantisation to ONNX reduced median CPU latency from 56.3 ms to 44.8 ms and file size from 6.3 MB to 3.4 MB for a loss of 0.007 mAP@50, measured over 20 benchmark images on an Apple M1 processor, giving roughly 22 frames per second without a graphics processor.

G. Negative results

Four failures constrain the design space and are reported because each was expected to succeed.

Firstly, open-vocabulary detection is unusable for this task. YOLO-World, prompted with ev battery module and busbar, attained 0.004 mAP@50 on the benchmark, battery internals being too far outside its training distribution.

Secondly, naive multi-source merging reduced module mAP@50 to 0.094 and aggregate accuracy to 0.331 on the single-source test split, below the 0.818 single-source baseline. More data made the model worse. This result motivated the screening procedure of Section IV-C.

Thirdly, automatic relabelling of the flagged source was attempted with an open-vocabulary grounding model in order to reconcile it with the consensus convention. On greyscale macro imagery the proposals were unusable, recovering two of approximately five modules on clean frames and producing boxes covering half the frame on close-ups. Convention conflicts are therefore not cheaply repairable after collection, which argues for agreeing annotation guidelines before collection rather than after.

Fourthly, copy-paste compositing combined with synthetic glare augmentation degraded the detector. Validation mAP@50 declined monotonically over the first six epochs, from 0.503 to 0.270, against a baseline of 0.818, and training was terminated. Both techniques are widely reported as beneficial elsewhere, and the likely cause here is label noise introduced at paste boundaries together with glare intensity that does not match the statistics of the real bright-condition imagery.

VI. Limitations

The 66% relative loss of Section V-A is measured against an in-domain benchmark that is itself the convention-divergent source. It therefore conflates generalisation loss with the specialist having been trained on an idiosyncratic convention, and should be read as an upper bound on the generalisation component; a second in-domain reference source would separate the two. Each configuration was trained once, so the 0.092 mAP@50 margin between RF-DETR and YOLO11n carries no confidence interval, and the two differ in epoch budget, resolution and optimiser because architecture defaults were used. That comparison is accordingly one between two configurations as commonly deployed, not a controlled comparison of architectures.

The benchmark inherits the annotation conventions of its sources, with no independently relabelled gold subset, so model error and label error cannot be separated on it; this is material for a study whose central claim concerns annotation quality. Both screens assume divergent sources are a minority, since the median defines the consensus, and the training-free index assumes divergence expressed as granularity. The corpus is assembled from public sources, so coverage is uneven and licences vary, one source at 24% of training images being CC BY-NC-SA 4.0 with a propagating ShareAlike condition. Finally, busbar accuracy of 0.35 is not sufficient for autonomous operation, and these results characterise the failure modes of the task rather than deliver a deployable detector.

VII. Conclusion

This paper presented a cross-facility benchmark for electric vehicle battery component detection. Single-source evaluation was shown to overstate deployment accuracy sufficiently to mislead, the same detector losing two thirds of its accuracy across pack sources. Annotation convention, rather than architecture, proved the dominant failure mode: the proposed screening procedure identified the divergent source from a single training run and recovered 0.227 module mAP@50, and a training-free variant flagged the same source with no training at all. A self-supervised pretrained backbone supplies robustness to annotation shift rather than improved detection of ordinary modules.

Taken together, these results indicate that dataset design, and specifically agreement on annotation convention before collection, is the first thing to get right in this application. Future work will construct an independently relabelled gold subset in order to separate model error from label error, and will extend the positive control to the model-based screen.

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